

Underwater Vision-Based Surveillance Using ML and AI

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Day: 4 Dataset Schema & Risk Analysis

Group:O2

1. Objective

The objective of Day 4 is to understand the structure, labels, targets and possible risks of all three supplied underwater datasets before starting model training.

2. Dataset Overview

Dataset	Classes	Total Images*	Structure
well.v8i.yolov8	4	4155	Train / Valid / Test
DIATAquarium.v4i.yolov8	11	9281	Train / Test
Aquatic Plant.v2i.yolov8	1	2147	Train / Valid / Test

Classes

well.v8i.yolov8 (4 classes):

- Inlet-pipe
- fishes
- school-of-fish
- stone

DIATAquarium.v4i.yolov8 (11 classes):

- Aquatic Plant
- Camera
- Conch-Shell
- Fish
- Fishes
- Pluco
- Shark
- Temperature Sensor

- Water-Pump
- Water-filter
- betta

Aquatic Plant.v2i.yolov8 (1 class):

- Aquatic_plant

3. Dataset Schema

All three datasets use the YOLO annotation format. Each image should have a corresponding .txt label file with the same filename.

Example:

image001.jpg image001.txt

YOLO labels contain:

class_id x_center y_center width height

Coordinates are normalized (0 to 1). The detection target is object class + bounding box.

4. Train / Validation / Test

Train: Used for model learning.

Validation: Used for model checking and tuning.

Test: Used for final evaluation, where available.

A correct split should preserve:

- Class distribution — each class should be represented proportionally across train, validation and test, not concentrated in one split.
- Video-clip integrity — since images are frames extracted from video, all frames from the same source recording should stay entirely within one split. No two splits should contain frames from the same clip.
- No duplicate or highly similar images appearing across different splits.

5. Dataset Risks

Risk	Description
Missing data	Images without a matching label file, or label files without a matching image.
Annotation validity	Empty label files, out-of-range coordinates, or wrongly assigned class IDs.
Class imbalance	Some classes (e.g. rare species/objects) may have far fewer labelled instances than others across all three datasets.
Temporal / video-frame leakage	Filenames (e.g. 20240102-20240102153559...mp4-1668_jpg...) show images were extracted as frames from continuous video recordings, not independent photos. Near-duplicate consecutive frames from the same clip could appear in both train and validation/test, letting the model see near-identical content in both — this must be checked before splitting.
Duplicate / near-duplicate images	A direct consequence of frame-based extraction — visually near-identical frames from the same recording.
Data leakage between splits	General risk of any overlap (exact or near-duplicate) between train, validation and test sets.
Image-quality variation	Frames include both grayscale IR/night-vision captures and color daylight captures, which represent different visual conditions the model must generalize across.
Incorrect paths in data.yaml	DIATAquarium's data.yaml currently points its validation path to test/images even though no separate Test folder exists in the dataset — see Section 6.

6. Risk Checklist before Training

- ☐ Verify image and label counts per split, per dataset
- ☐ Check image-label filename matching (no orphan images or labels)
- ☐ Check annotation validity (no empty/out-of-range/misassigned labels)
- ☐ Check class distribution across train/valid/test
- ☐ Scan filenames for repeated video-source IDs across splits (temporal leakage check)
- ☐ Check for duplicate/near-duplicate images

- ❑ Verify data.yaml paths against actual folder structure